

ML_ZoomCamp_Homework1

```
import numpy as np
import pandas as pd

df = pd.read_csv("car_fuel_efficiency.csv")

df.head(5)
```

	engine_displacement	num_cylinders	horsepower	vehicle_weight	\
0	170	3.0	159.0	3413.433759	
1	130	5.0	97.0	3149.664934	
2	170	NaN	78.0	3079.038997	
3	220	4.0	NaN	2542.392402	
4	210	1.0	140.0	3460.870990	

	acceleration	model_year	origin	fuel_type	drivetrain
0	17.7	2003	Europe	Gasoline	All-wheel drive
1	17.8	2007	USA	Gasoline	Front-wheel drive
2	15.1	2018	Europe	Gasoline	Front-wheel drive
3	20.2	2009	USA	Diesel	All-wheel drive
4	14.4	2009	Europe	Gasoline	All-wheel drive

	fuel_efficiency_mpg
0	13.231729
1	13.688217
2	14.246341
3	16.912736
4	12.488369

Question 1. Pandas version (1 point)

```
print(pd.__version__)
```

2.2.3

Question 2. Records count (1 point)

```
records_count = df.shape[0]
print("Q2. Records count:", records_count)
```

Q2. Records count: 9704

Question 3. Fuel types (1 point)

```
fuel_types_count = df["fuel_type"].nunique()
print("Q3. Fuel types:", fuel_types_count)
```

Q3. Fuel types: 2

Question 4. Missing Values (1 point)

```
missing_values = (df.isnull().sum() > 0).sum()
print("Q4. Missing values:", missing_values)
```

Q4. Missing values: 4

Question 5. Max fuel efficiency (1 point)

```
max_fuel_eff = df["fuel_efficiency_mpg"].max()
print("Q5. Max fuel efficiency:", max_fuel_eff)
```

Q5. Max fuel efficiency: 25.96722205

Question 6. Median value of horsepower (1 point)

```
median_hp_earliest = df[df["model_year"] == df["model_year"].min()]
["horsepower"].median()
median_hp_latest = df[df["model_year"] == df["model_year"].max()]
["horsepower"].median()
```

```
if median_hp_latest > median_hp_earliest:
    hp_change = "Yes, it increased"
elif median_hp_latest < median_hp_earliest:
    hp_change = "Yes, it decreased"
else:
    hp_change = "No"
```

```
print("Q6. Median value of horsepower:", hp_change)
```

Q6. Median value of horsepower: Yes, it increased

Question 7. Sum of weights (1 point)

```
sum_weights = df["vehicle_weight"].sum()
print("Q7. Sum of weights:", sum_weights)
```

```
scaled_sum = sum_weights / 1e6
print("Scaled sum (in millions):", scaled_sum)
```

Q7. Sum of weights: 29124430.759705596
Scaled sum (in millions): 29.124430759705596